

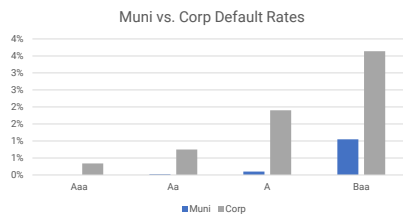
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Municipal Strategy | North America

Municipal Market Primer

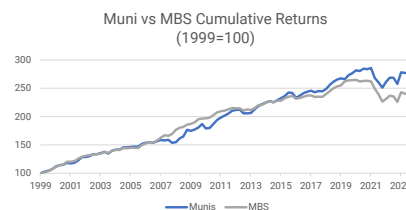
Munis build America's infrastructure, large and small. Muni bonds offer investors exceptional credit quality, low volatility, and predictable tax-free income. If you've driven, flown, gone to school, or turned on a tap – chances are munis made it happen.

Exhibit 1: Munis default less than most similarly rated credit markets



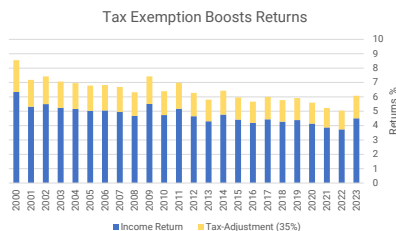
Source: Moody's, Morgan Stanley Research

Exhibit 2: Muni returns meet or exceed other callable & negatively convex asset classes



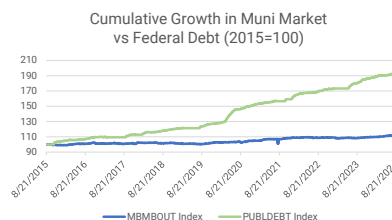
Source: Bloomberg, Morgan Stanley Research

Exhibit 3: The tax exemption provides a valuable boost not captured in returns statistics



Source: ICE, Morgan Stanley Research

Exhibit 4: Unlike the federal government, munis have shown remarkable discipline in total debt issuance



Source: Bloomberg, Morgan Stanley Research

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We thank Hitesh Garg, Municipal Strategy, for his contributions to this report.

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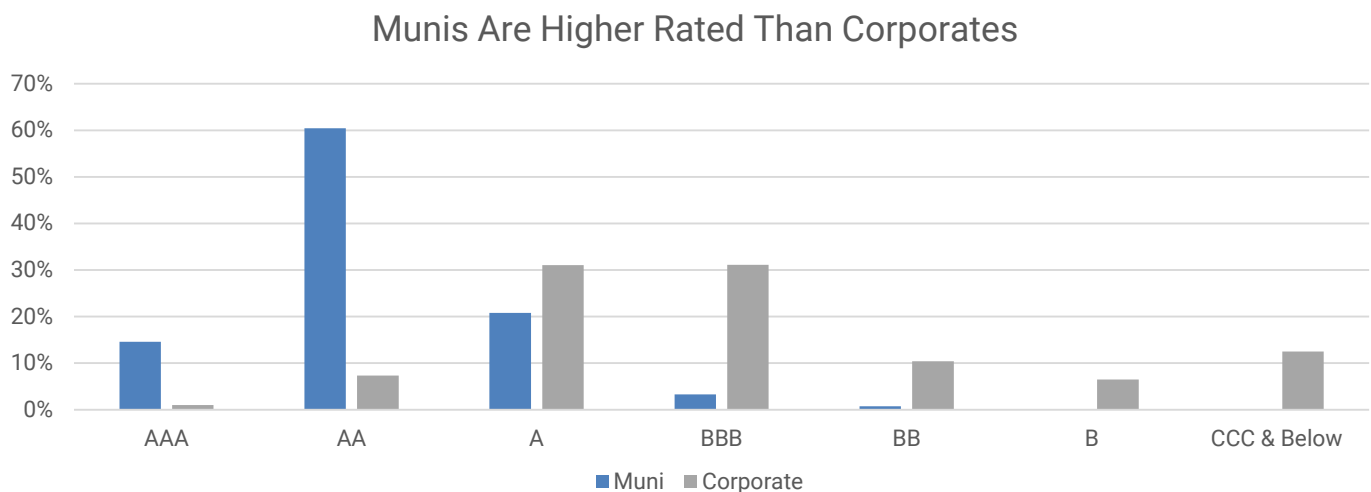
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Why Invest in Munis

Tax-free income

Municipal bond interest income is usually exempt from federal, state, and local taxes. Investors in a high tax bracket would need to find a bond yielding 1.5-1.9x more than a muni for an equivalent after-tax return.

Exhibit 55: Munis Benefit From Much Higher Ratings Than Corporates



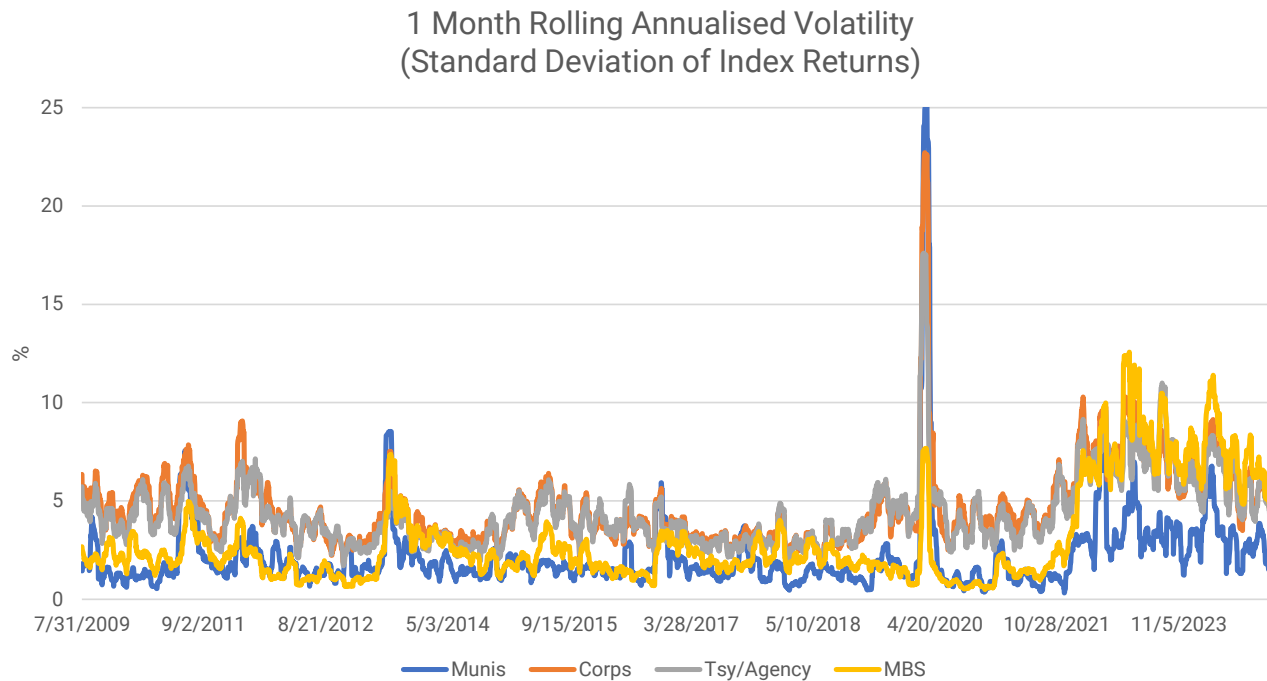
Source: Bloomberg, Morgan Stanley Research. Note: for bonds with a Bloomberg composite rating.

Very high credit quality

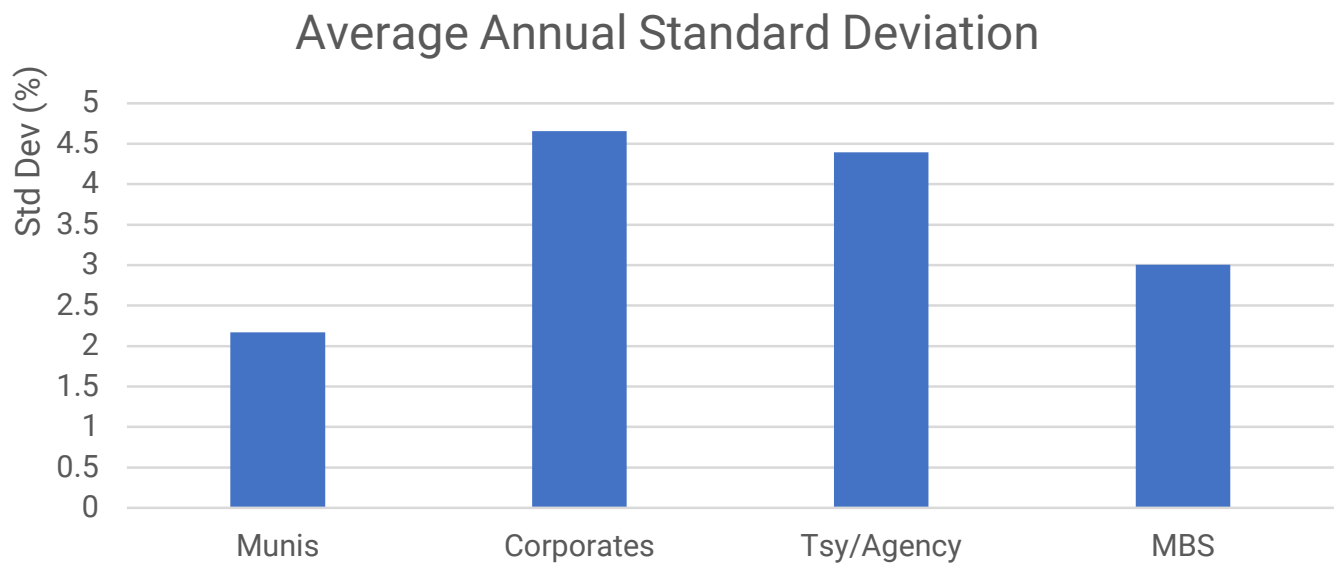
The taxing power of munis is more dependable than the pricing power of any private company. Municipal bonds default and are downgraded at a substantially lower frequency than corporate bonds.

Lower volatility

Munis are mostly held by households, who typically trade infrequently. As a result, muni returns are less volatile than corporates or Treasuries.

Exhibit 6: Rolling Volatility

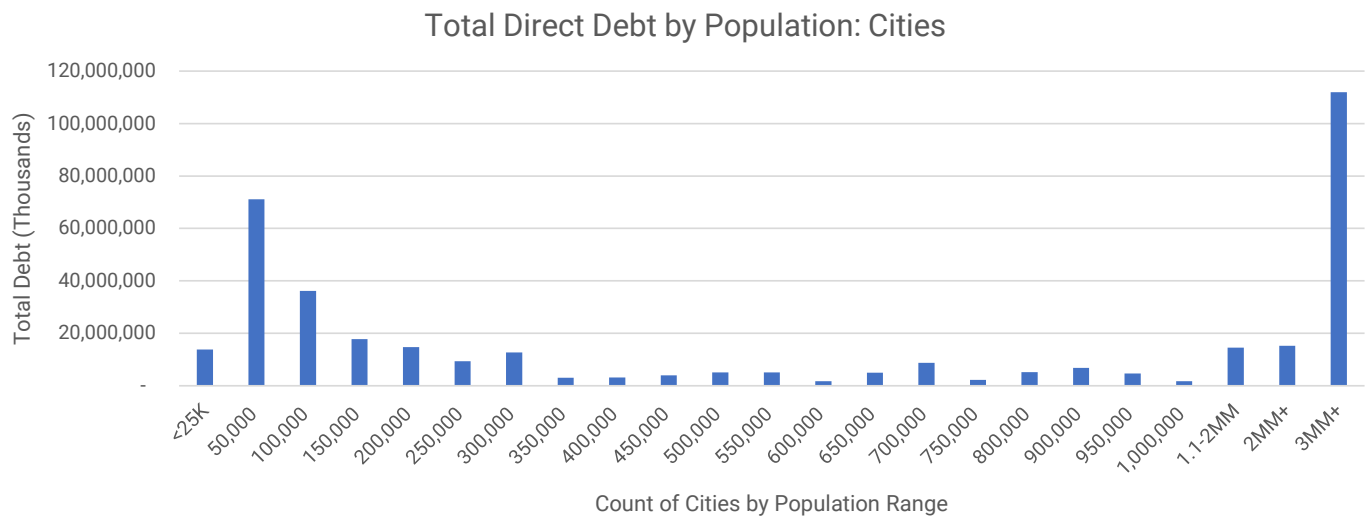
Source: ICE, Morgan Stanley Research

Exhibit 7: Average Volatility

Source: ICE, Morgan Stanley Research

Why Issue Munis

Exhibit 88: Munis let smaller governments (with less than 50K population) borrow for 30Y+



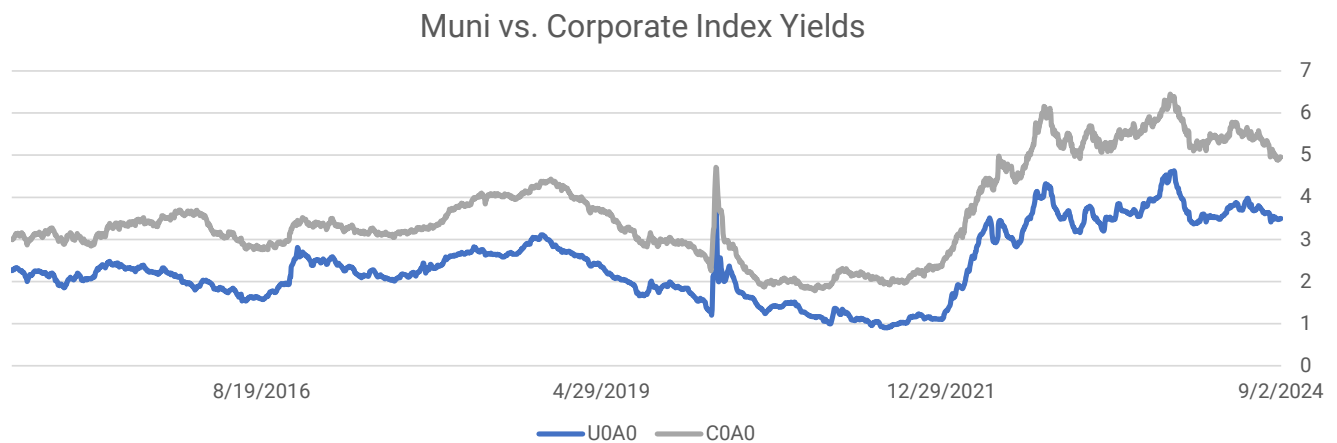
Source: InvestorTools, Morgan Stanley Research

Lock-in 30Y+ terms at low rates

The municipal bond market provides financing that matches a project's useful life.

Munis allow issuers extended terms unavailable in the bank loan market. The tax exemption also allows municipalities to borrow at substantially lower interest rates than companies.

Exhibit 109: Muni vs. Corporate Index Yields



Source: ICE, Morgan Stanley Research

Open to issuers large and small

Munis make small governments less dependent on local banks or state authorities.

Passive, "hands-off" investor base

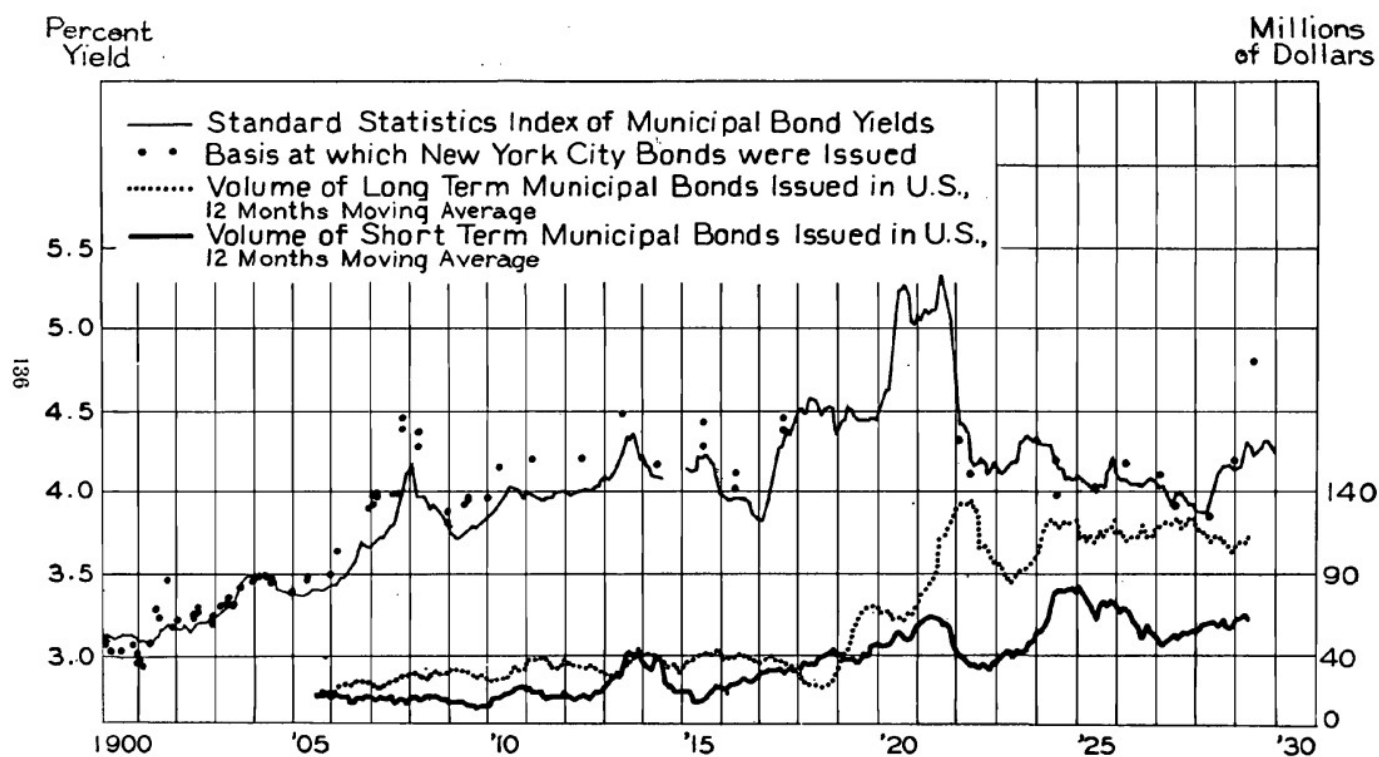
Municipal bond investors are hands-off. Muni issuers rarely face the same level of scrutiny demanded by investors in public companies.

Tried & tested market

Local governments have regularly issued municipal bonds for over 150 years. The first municipal bonds in the US were issued over 200 years ago.

Exhibit 10: Reliable statistics on municipal bond yields go back to 1900

CHART 27.—MUNICIPAL BOND YIELDS AND MUNICIPAL BOND ISSUES IN THE UNITED STATES, 1900–1929.



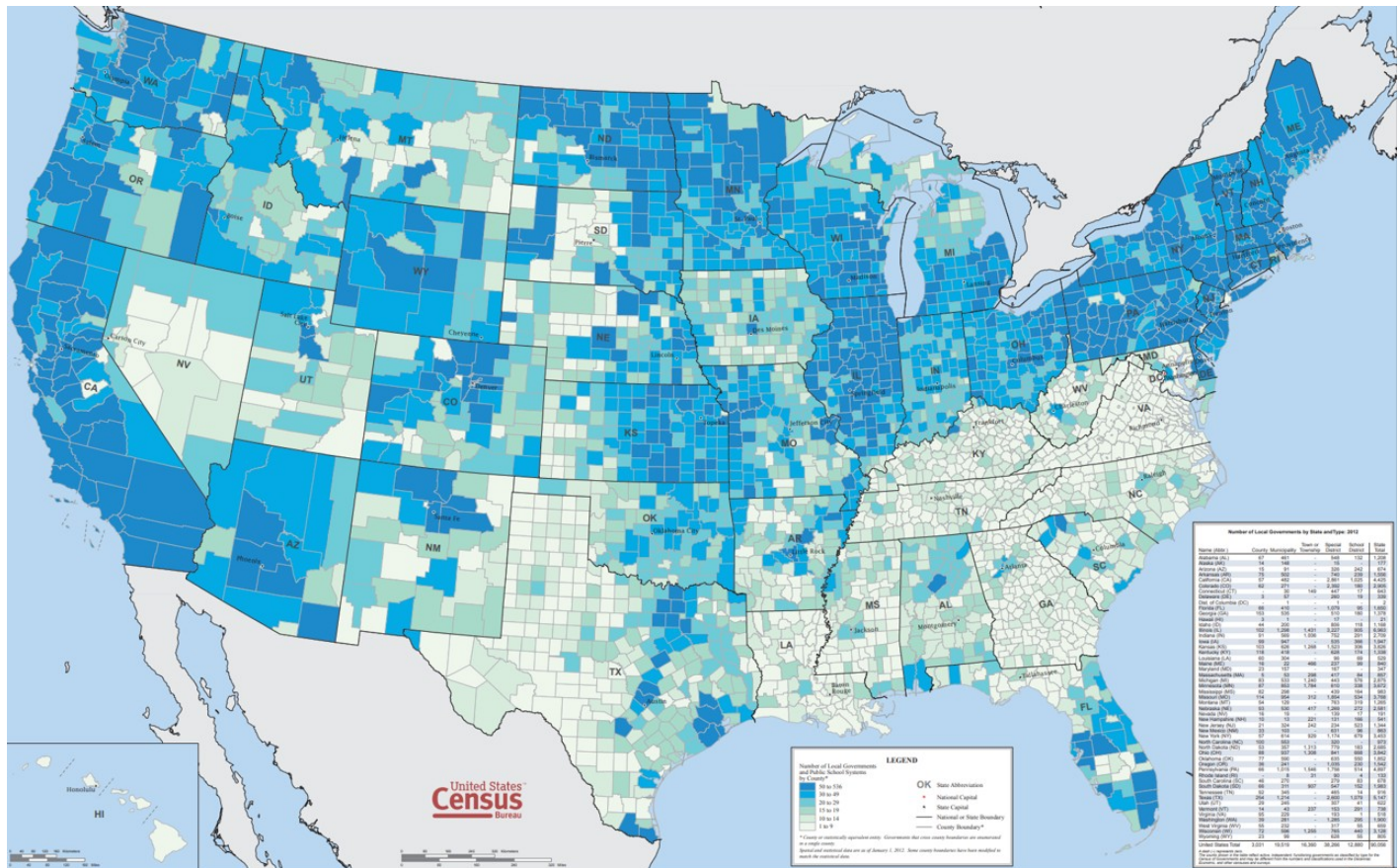
Source: Leo Wolman, "The Financing of Public Construction," NBER, 1930

What Munis Build

Government infrastructure

State and local governments finance the majority of America's infrastructure:

Number of local governments per county (dark blue = >50 governments per county)



Source: US Census Bureau's 2012 Census of Governments

Not-for-profit facilities

Hospitals and universities issue tax-exempt bonds. To access tax-exempt financing, they borrow through a government conduit agency.

Affordable housing

Many first-time homebuyers benefit from state housing finance agencies (HFAs). State HFAs use municipal bonds to underwrite affordable mortgages on single- and multi-family properties.

Industrial & private facilities

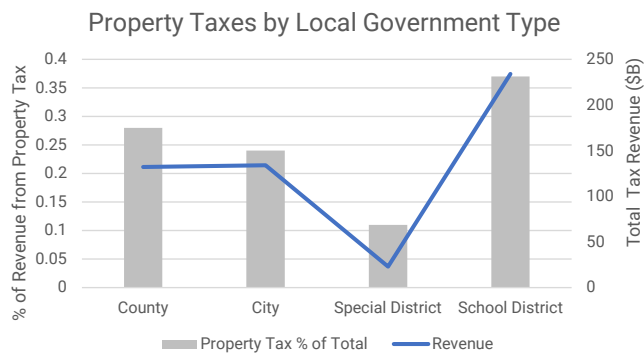
Various kinds of transportation projects are tax-exempt, even if financed by private developers. Airports are a common venue for public-private partnerships.

Who Issues Munis

General governments

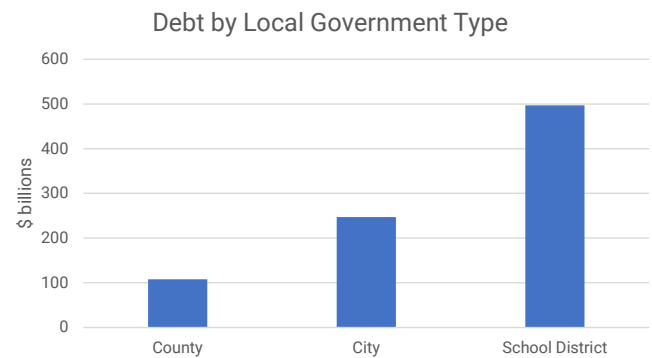
Cities, counties, and school districts all issue unlimited general obligation (unlimited GO) debt. They typically have independent taxing authority, though state law may limit to what degree taxes can change from year to year.

Exhibit 11: Local governments' dependence on property tax varies



Source: Tax Policy Center, Morgan Stanley Research

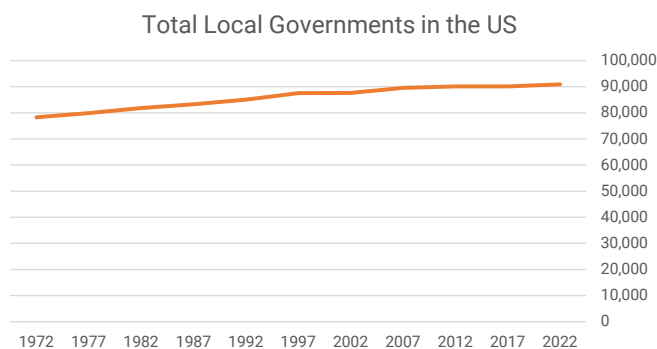
Exhibit 12: School districts are one of the largest kinds of local government GO issuers



Source: Bloomberg, Morgan Stanley Research

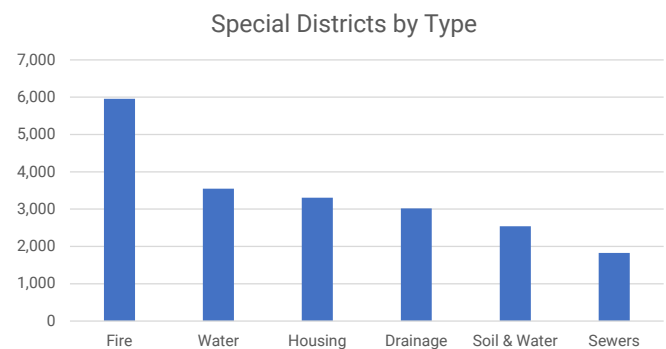
Limited-purpose governments

Exhibit 13: There are about 90,000 local governments in the US



Source: US Census Bureau, Morgan Stanley Research

Exhibit 14: Most special districts focus on a specific municipal responsibility



Source: US Census Bureau, Morgan Stanley Research

Private not-for-profits

Hospitals and universities issue tax-exempt bonds. To access tax-exempt financing, they borrow through a government conduit agency. Many not-for-profits have substantial financial resources; the largest universities and hospitals have credit ratings which exceed those of many states and cities.

Munis & Asset Allocation

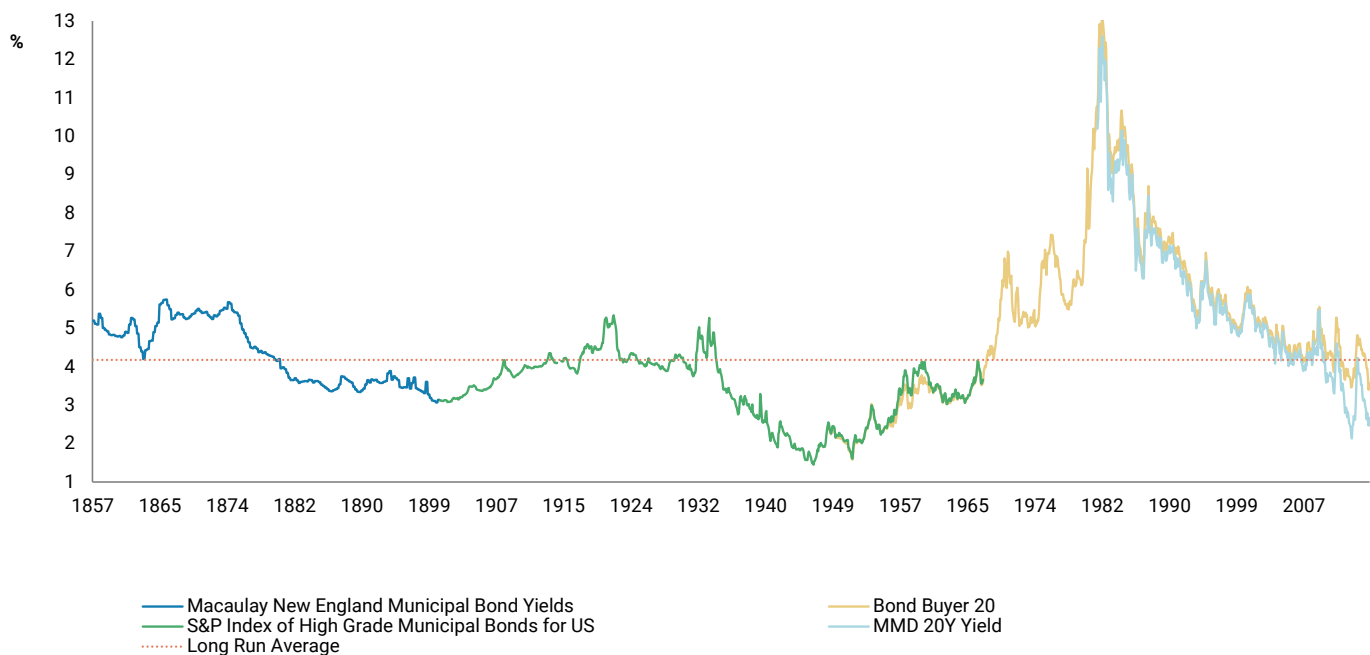
The cost of waiting

- Waiting for a better entry point by sitting in cash can seem like the "safe" choice, but it actually amounts to a strong market timing call.
- There is little evidence to suggest that interest rates return to a long-run mean over the short to medium term.
- Neutral-duration portfolios tend to outperform short-duration portfolios over time.

Over the very long run, muni yields have been lower than you think

Since the 1850s, muni yields have averaged a little over 4%:

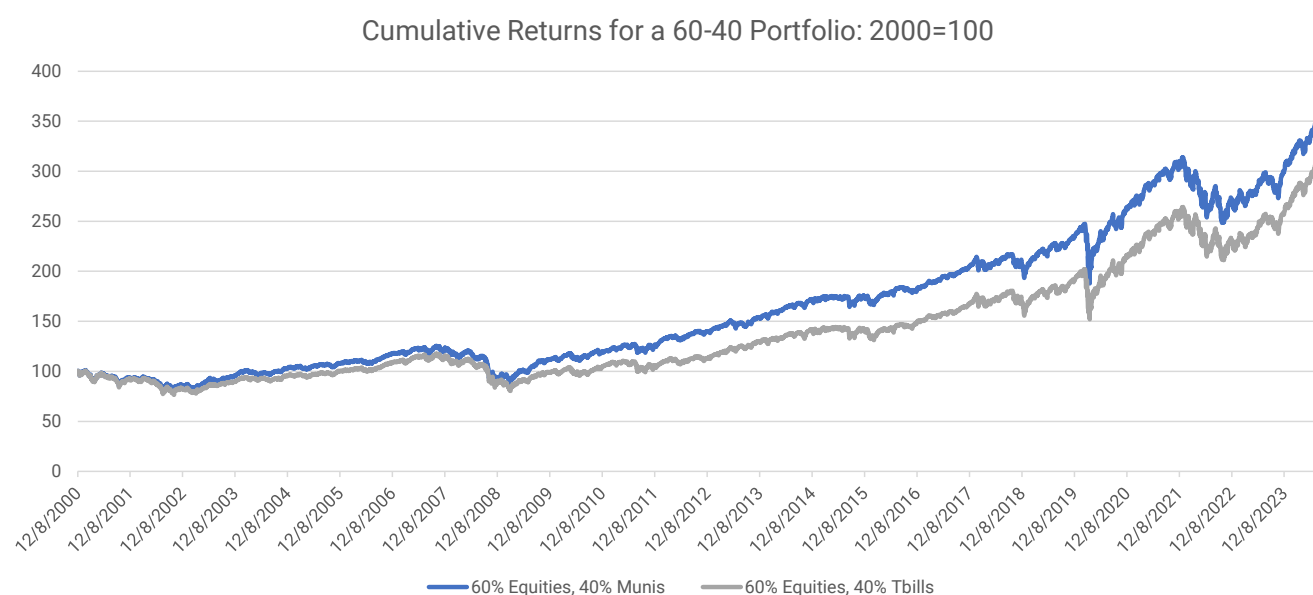
Exhibit 15: Except for the 1970s and 1980s, muni yields were never above 6%



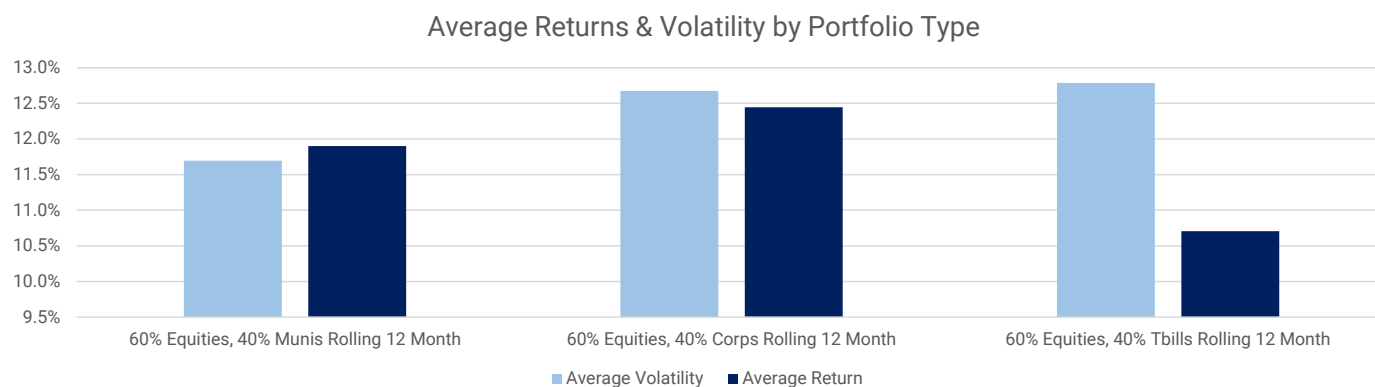
Source: S&P, Bond Buyer, St Louis Fed, MMD, Morgan Stanley Research

Munis diversify better than cash

Historically, a portfolio of stocks and munis has outperformed a portfolio of stocks and money markets.

Exhibit 16: A stocks/munis portfolio outperformed stocks/cash

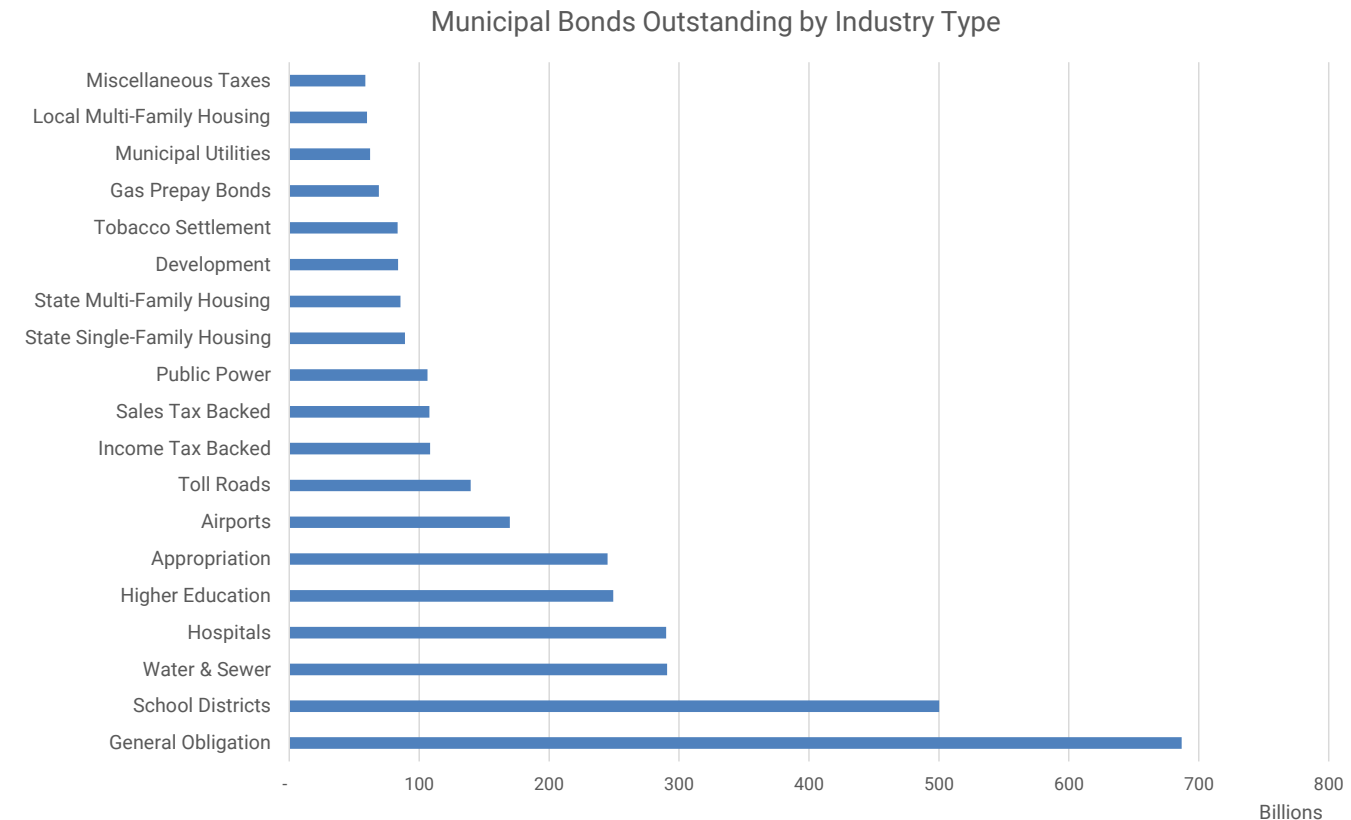
Source: Bloomberg, Morgan Stanley Research

Exhibit 17: Returns & Volatility by Portfolio Type

Source: Bloomberg, Morgan Stanley Research

How Munis Are Repaid

Exhibit 18: Outstanding bonds by municipal industry/sector



Source: Bloomberg, Morgan Stanley Research

General obligation pledge

The "full faith and credit" pledge of local governments often works like a senior unsecured pledge. In some states, laws create a statutory lien from general obligation debt. However, this is the exception and not the rule.

Typically, property taxes back general obligation debt.

Revenue pledge (governmental tax)

Governments may identify specific taxes to repay bondholders. Sales taxes, transportation taxes, and income taxes are often pledged. Typically, there is no recourse to the government if the pledged revenues fall short.

Revenue pledge (business-like enterprises)

Government-owned utilities typically pledge to set rates at levels sufficient to meet a minimum debt service coverage ratio. The pledge of revenues may be either gross or net of operating expenses.

Appropriation debt

Many states have laws that require voter approval for general obligation debt. To address this, many governments instead issue debt for which debt service payments must be appropriated each year. A failure to appropriate is typically not an event of default. Appropriation debt is always subordinate to general obligation debt, in the way that a senior unsecured pledge ranks well above preferred stock.

Hospitals & universities

Hospital and universities make debt service payments based on their free cash flow. Typically, specific assets (like endowments) are not specifically pledged.

Housing revenues

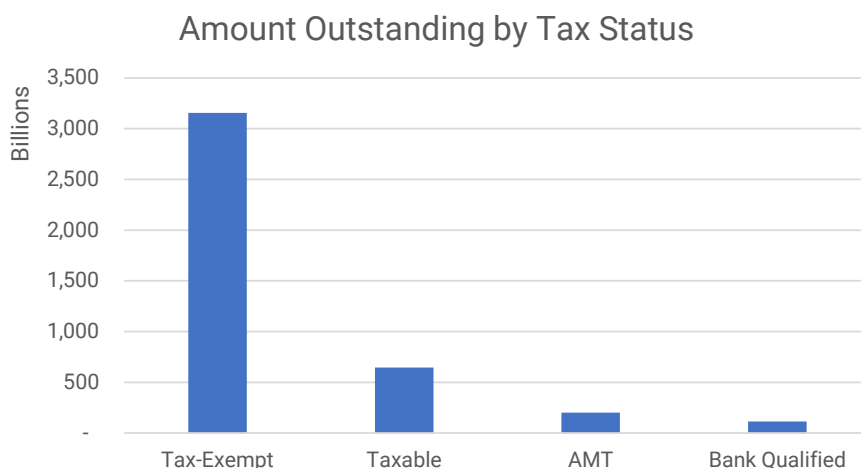
Housing finance agencies issue bonds that are repaid by mortgage payments, as with other kinds of mortgage-backed securities.

How Taxes Work (for Investors)

For most bonds, interest is tax-free

Munis that finance government-owned facilities are typically tax-free. Current refinancings are also tax-free; early refis (known as advance refundings) are not.

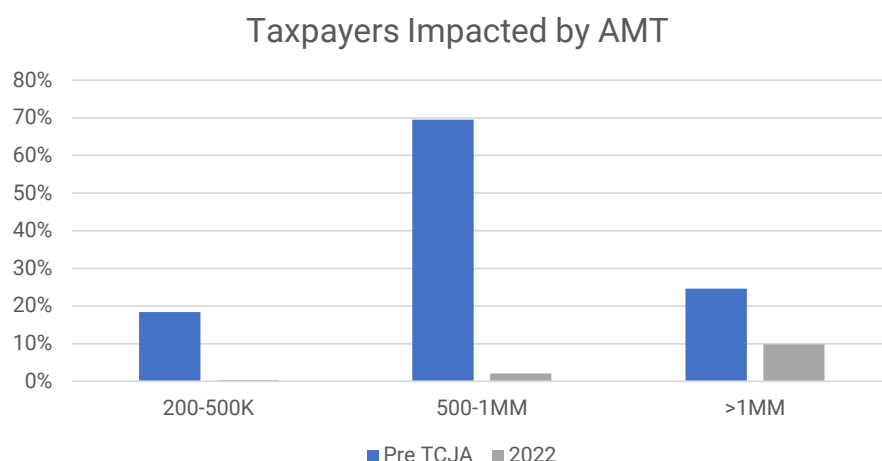
Exhibit 19: Most munis are federally tax-exempt



Source: Bloomberg, Morgan Stanley Research

Some bonds are subject to alternative minimum tax (AMT)

Exhibit 20: The Tax Cuts & Jobs Act (TCJA) reduced the burden of the AMT, though AMT provisions sunset in future years



Source: Tax Policy Center, Morgan Stanley Research

Muni projects that finance private infrastructure projects are often subject to the alternative minimum tax. The AMT aims to ensure that households with high levels of tax-advantaged income still pay some tax. Because of added complexity, investors have historically demanded a premium to own AMT bonds.

Some bonds are federally taxable

If a bond pays for an early refi, or if it will invest in financial securities (like for a pension), then it is typically federally taxable.

Exhibit 21: Taxable Muni Index Yields

Taxable Muni Yields Typically in-line with AAA-A rated Corporates



Source: ICE, Morgan Stanley Research

Exhibit 22: S&P Taxable Muni Index

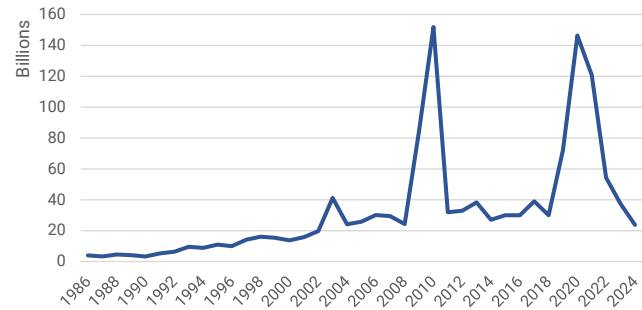
S&P Taxable Muni Index: Total Outstanding



Source: S&P, Morgan Stanley Research

Exhibit 23: Taxable Issuance by Year

Taxable Issuance by Year



Source: Bond Buyer, Morgan Stanley Research

All bonds are subject to capital gains

Though interest income may be tax-exempt, capital gains and losses are fully taxable.

Deeply-discounted bonds may incur additional tax

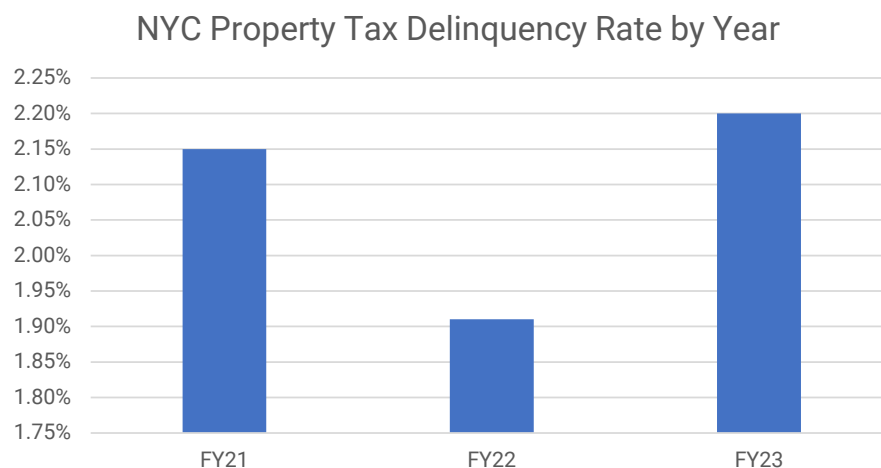
Bonds trading at a deep discount to their original issue price may be subject to a mix of ordinary income and capital gains taxes.

How Taxes Work (for Issuers)

Taxes are almost always paid

Property tax delinquencies remain low, even during recessions. In New York City, for instance, delinquencies remained under 3% even in the wake of COVID.

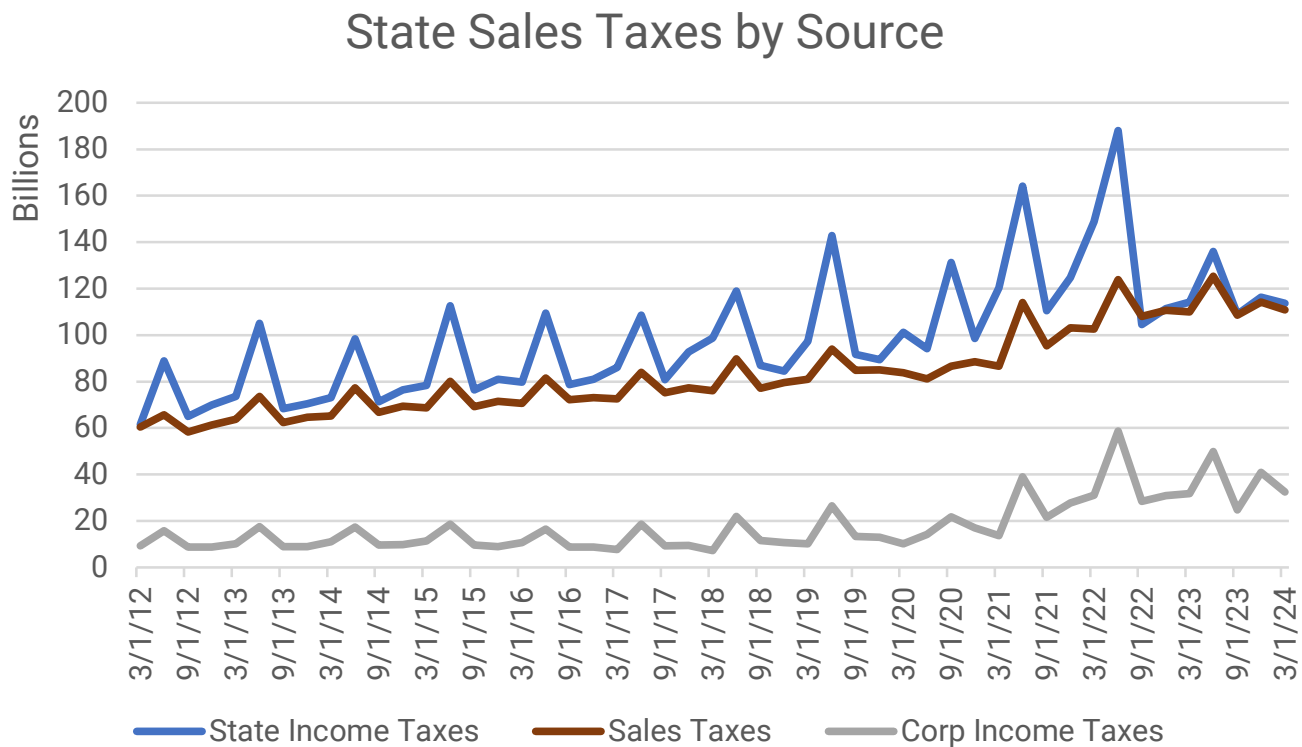
Exhibit 24: NYC property tax levy: about 98% paid each year



Source: [New York City Comptroller](#), Morgan Stanley Research

Some taxes are more volatile than others

Of large tax sources, income taxes are generally more volatile than sales taxes. Corporate income taxes are more unpredictable than individual income taxes.

Exhibit 25: Income taxes have both the most volatility and the most seasonality

Source: Bloomberg, US Census Bureau, Morgan Stanley Research

State governments share tax collections with local governments

Taxes are shared with local governments. School districts, for example, receive substantial funding from the state. Typically, these intergovernmental relationships are a source of credit quality.

Larger government units (like counties) often collect taxes on behalf of smaller governments (like school districts)

Small local governments typically do not collect taxes themselves; instead, they rely on a county officer.

Who Owns Munis

Mutual funds, closed-end funds, & ETFs

Open-end mutual funds, closed-end funds, and ETFs hold over \$1 trillion in munis. ETFs are gaining share at the expense of mutual funds. Closed-end funds have been stagnant for many years.

Separately-managed accounts

Aside from ETFs, separate accounts are the fastest-growing channel. They are on pace to overtake mutual funds as the largest professional holder. Estimates for SMA AUM vary.

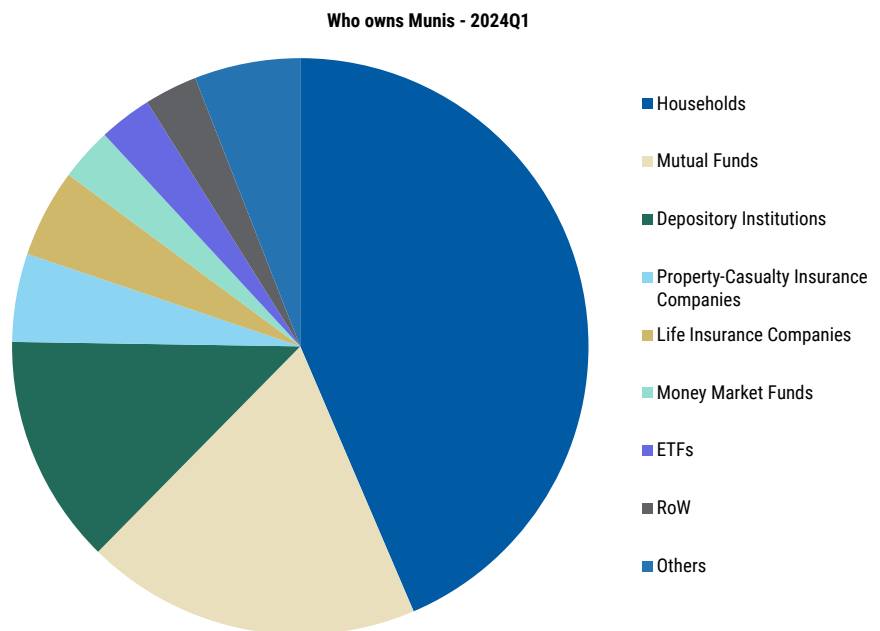
Banks & insurance companies

Banks and insurance companies used to be the largest owners of munis. But several decades of corporate tax cuts and tax reform reduced their attractiveness to companies' bottom line.

International investors

International investors are a growing presence in the muni market. They almost exclusively invest in taxable municipals.

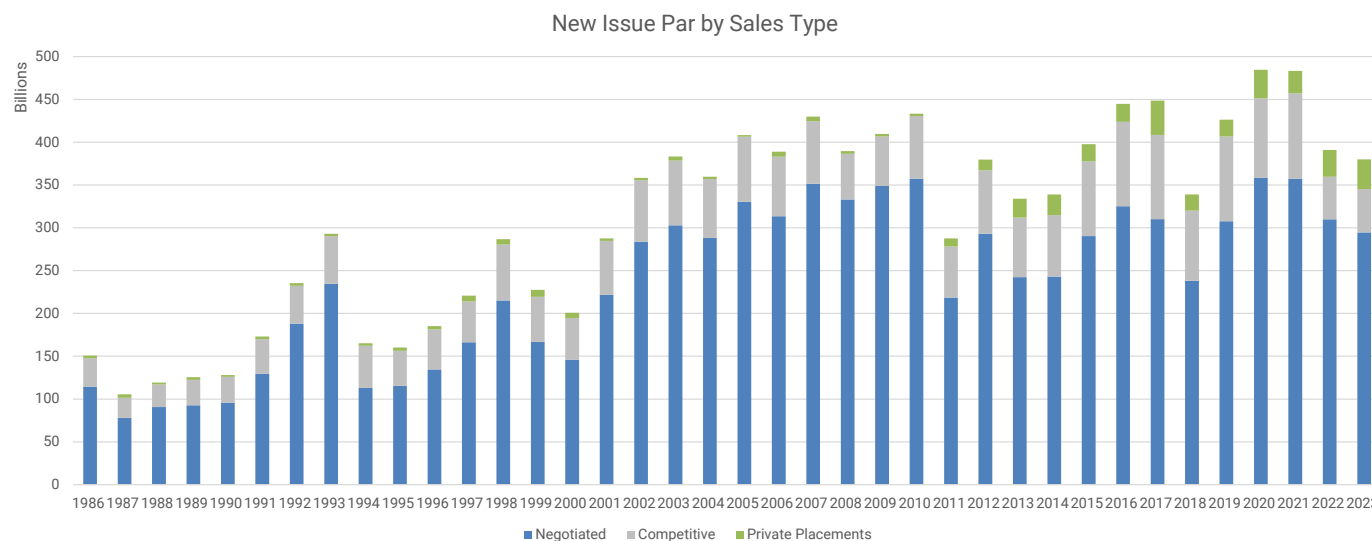
Exhibit 26: Households own two-thirds of the market



Source: Federal Reserve, Morgan Stanley Research

How Bonds Are Issued

Exhibit 27: Total sales by type



Source: Bloomberg, Morgan Stanley Research

Negotiated sales

A negotiated sale is akin to a traditional underwriting process. About half of issuers come to market in this way.

Competitive sales

Competitive sales let bidders structure deals as they wish, subject to parameters set forth by the issuer. The highest bidder wins the deal, defined typically as the lowest true interest cost.

Direct placements & bank loans

A direct placement or bank loan may offer faster time to issuance. For some issuers, it can also simplify and minimize required disclosure.

How Disclosure Works

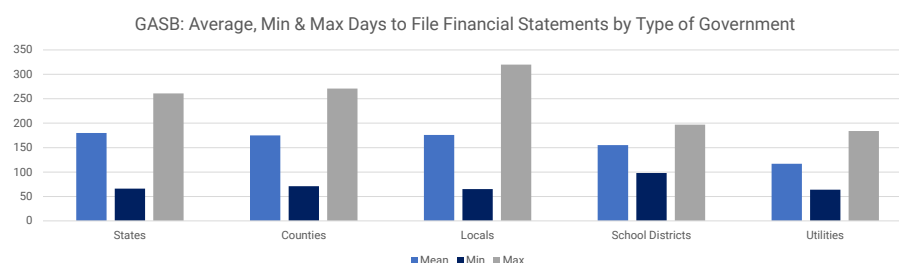
Governments are not subject to reporting requirements

SEC rule 15c2-12 requires that dealers ensure their issuer clients *agree* to provide disclosure on a regular basis. However, there is no *requirement* to do so.

Annual reports take longer to produce

An annual comprehensive report usually contains more information than a 10-K. However, it takes longer to produce.

Exhibit 28: Governments take 6-9 months to produce financial reports



Source: GASB, Morgan Stanley Research

Quarterly disclosure limited to healthcare & transportation enterprises

Hospitals reliably produce quarterly financial disclosure. Many large transportation agencies also produce limited quarterly statements detailing tolls, passengers, or other activity measures.

Government accounting uses a modified accrual standard

Governments have a separate set of GAAP standards. Their financial statements employ a modified accrual approach, which blends full accrual with cash basis accounting. In addition, not all issuers prepare GAAP statements.

Statements for business-type activities (like transportation entities) are often prepared on a full-accrual basis.

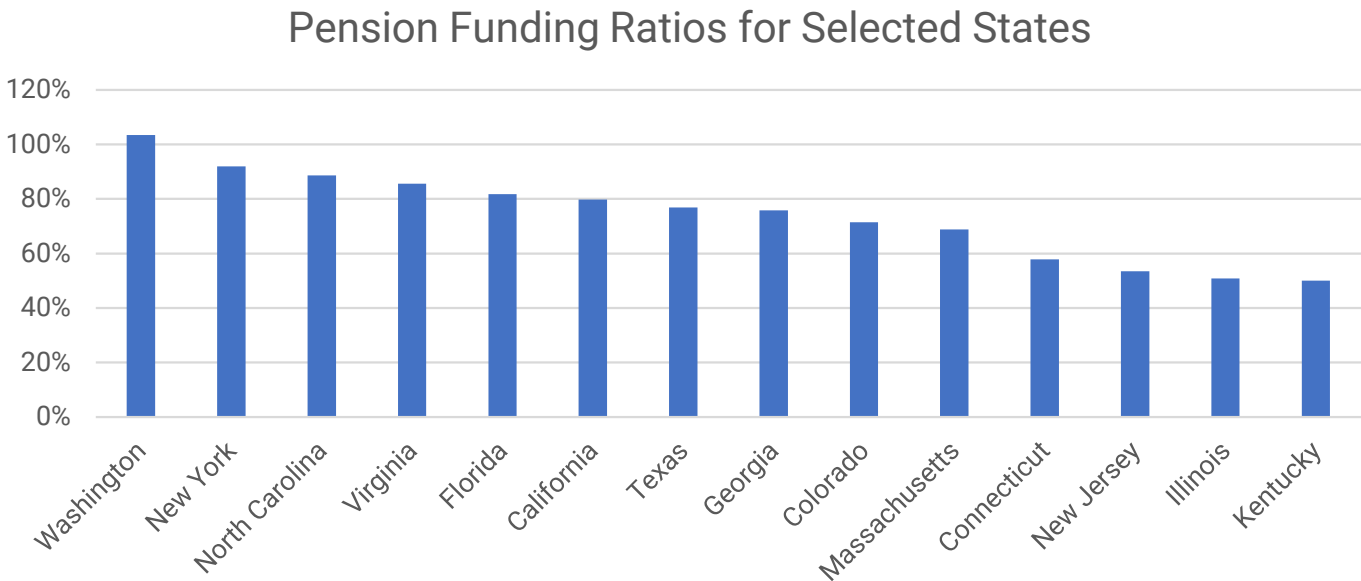
Budgets are balanced & calculated on a cash basis

Most governments propose a balanced budget each year. Typically, this is done on a cash basis for the general fund.

Public pension reporting is different

Public pension funds produce separate financial statements. In addition, government accounting requires estimates of funding ratios given changes in discount rates. Governments also calculate estimates of retiree healthcare liabilities (OPEB). Unlike for companies who participate in pension insurance programs like the PBGC, governments incur no penalties for carrying a net pension liability.

Exhibit 29: Most pensions are 70%+ funded



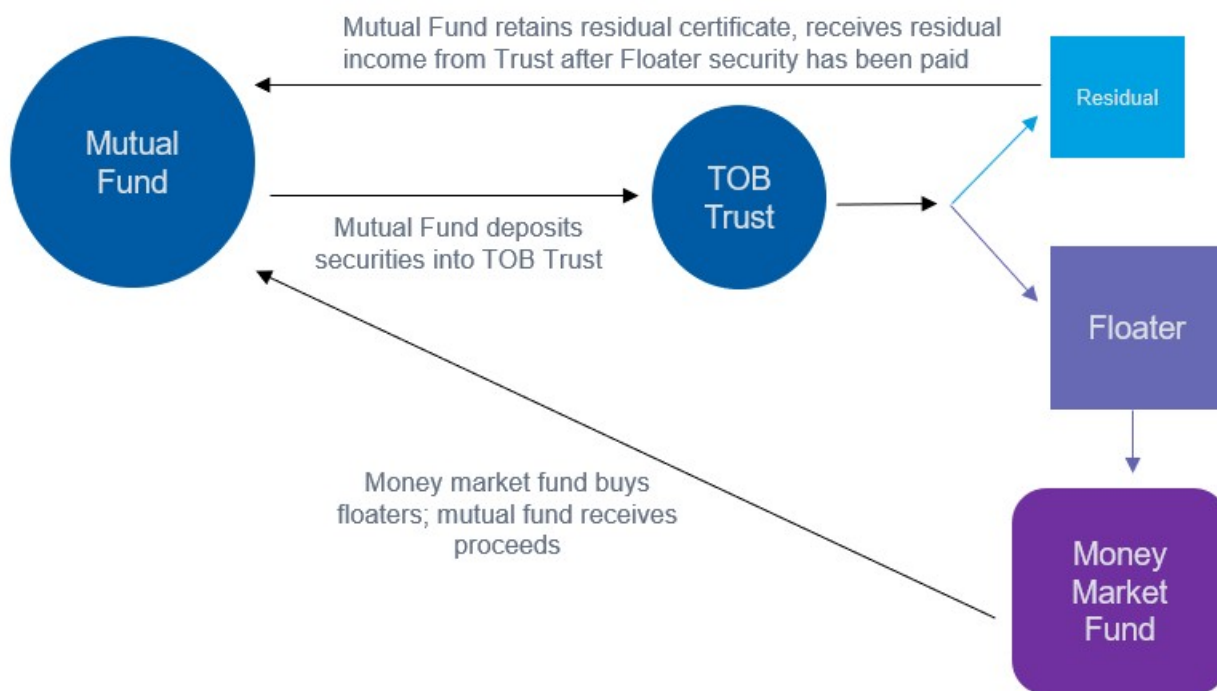
Source: Equable Institute, Morgan Stanley Research

How to Hedge & Add Leverage

Tender option bonds are most common way to lever

A tender option bond is an A-B trust that pairs a floating rate certificate with a single residual class. The structure preserves the tax exemption of the underlying bonds in the trust. TOB and other floating rate securities typically reset off the SIFMA rate, a weekly yield that averages around 67% of fed funds.

Simplified TOB diagram



Morgan Stanley Research

Rate locks are a growing hedge

Rate locks allow investors to express a view on the direction of the MMD or other similar AAA, general obligation yield curves.

A lack of tradeable credit derivatives

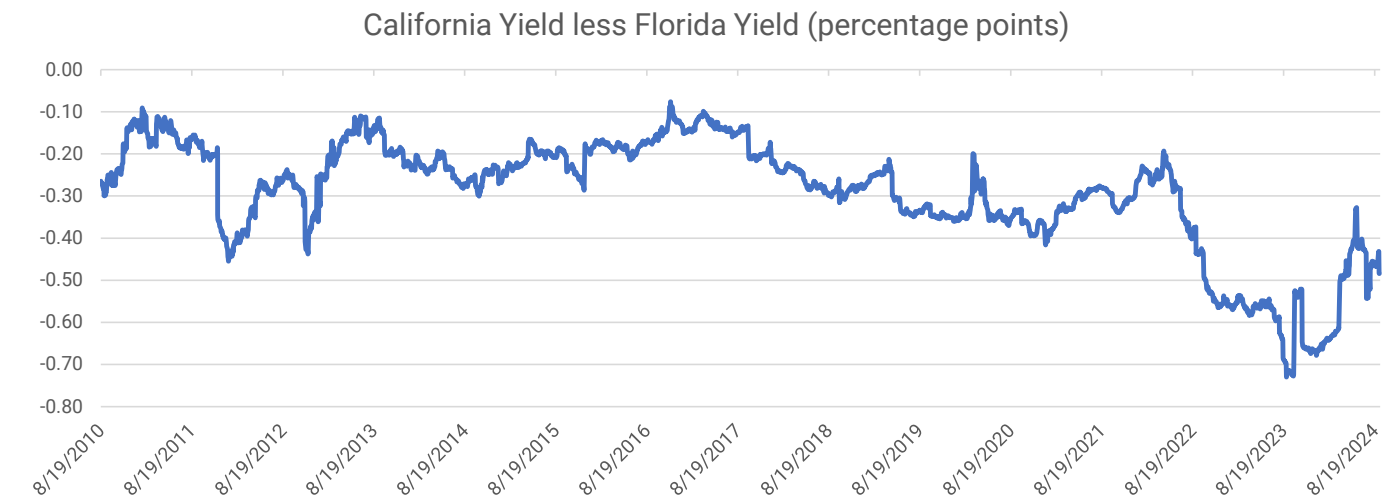
Because munis rarely default, there is no longer a liquid market in muni credit derivatives. MCDX rolls ceased several years ago.

How to Think About Spreads

Tax rates can impact state spreads

High tax states (like California) often trade tighter than states with no income tax (like Florida), even though Florida has a better credit rating.

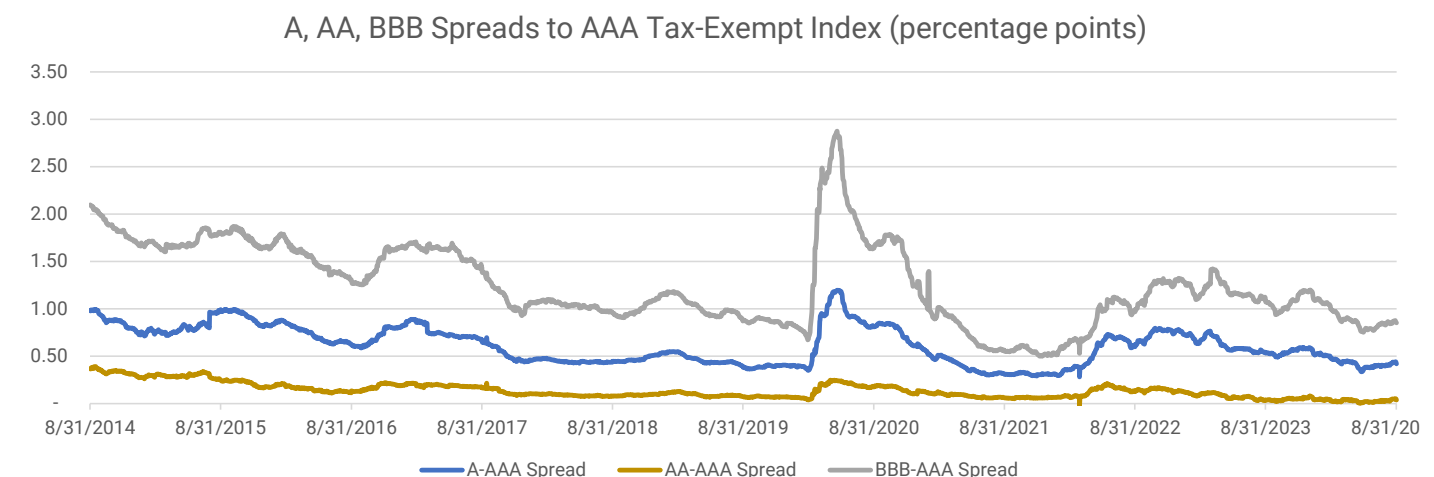
Exhibit 30: California bonds usually trade tight to Florida, given California tax exemption is worth more



Source: S&P Indices, Morgan Stanley Research

Tax-exempt spreads by rating

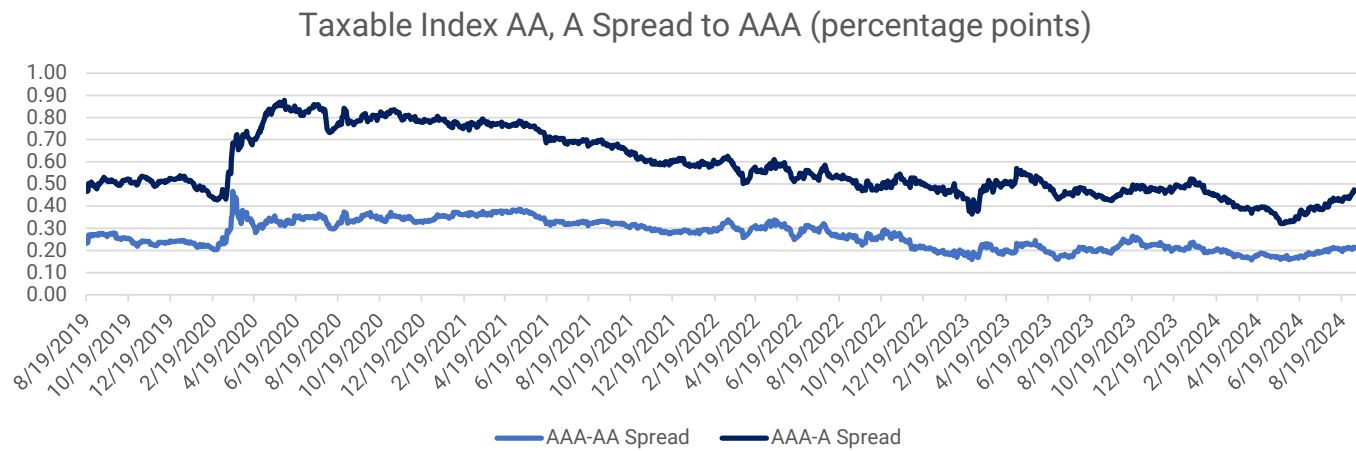
Exhibit 31: Long-run tax-exempt index spreads to AAA index



Source: ICE, Morgan Stanley Research

Taxable spreads by rating

Exhibit 32: Taxable Index Spreads to AAA



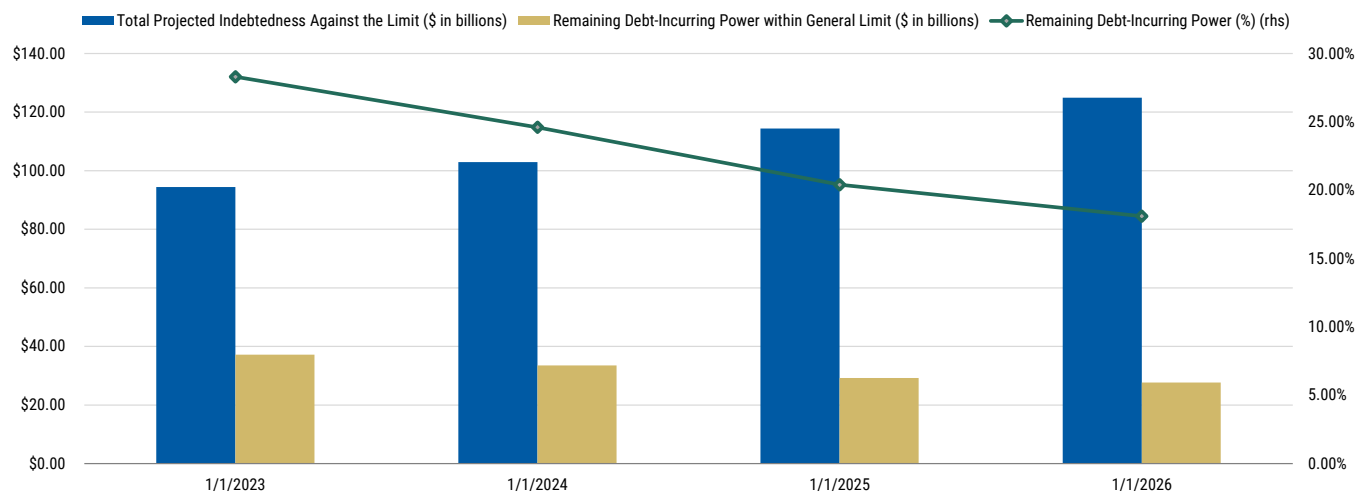
Source: S&P, Morgan Stanley Research

A Few Tips on Credit Analysis

Look at longer-term projections for debt capacity

Large issuers typically release a multi-year capital plan that includes issuance projections. State laws regulating issuance (debt limits) can help put a ceiling on forecasts.

Exhibit 33: A sample 'debt-incurring power' projection

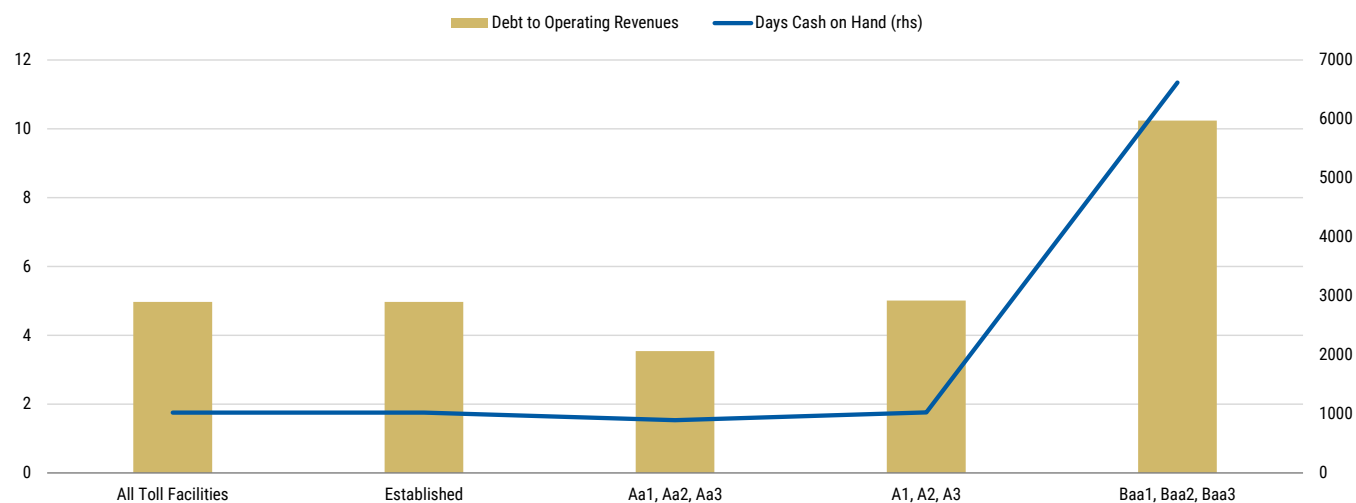


Source: New York City Comptroller, Morgan Stanley Research

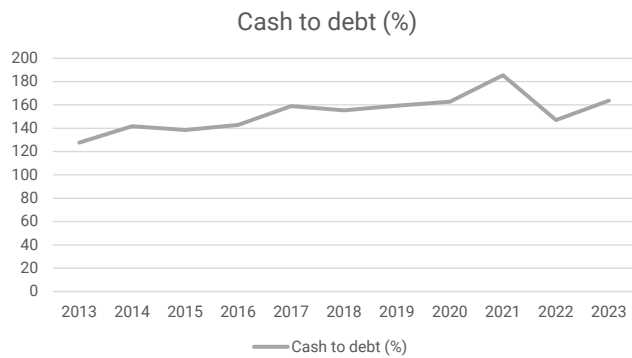
Transportation names typically have high days cash & high leverage

Many muni credits have very low EBITDA volatility. Infrastructure assets typically have low operating expenses. Combined, this means issuers generally have high DCOH and high debt-to-revenue.

Exhibit 34: Toll Roads: Selected Metrics



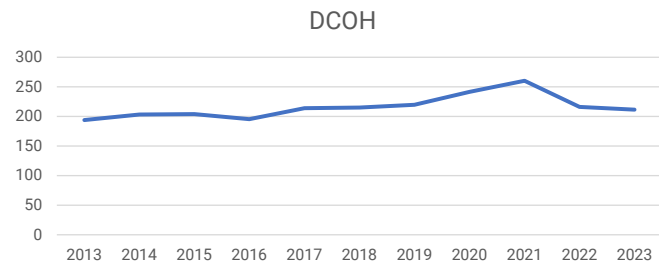
Source: Moody's, Morgan Stanley Research

Exhibit 35: Median cash to debt: NFP Hospitals

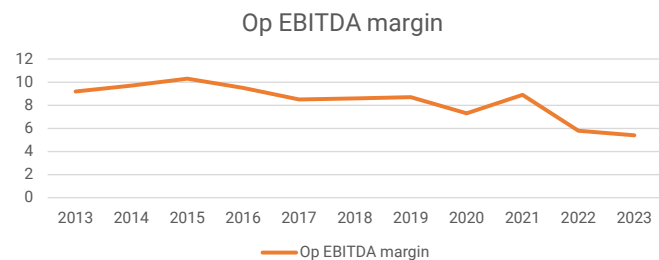
Source: S&P, Morgan Stanley Research

Healthcare: Balance sheet quality, EBITDA margins key

Although healthcare margins are volatile, not-for-profit hospital balance sheets are very strong. Since hospitals are a labor-intensive business with high operating expenses, analysts look to days cash on hand. They also pay attention to trends in operating EBITDA margin.

Exhibit 36: Hospital median DCOH

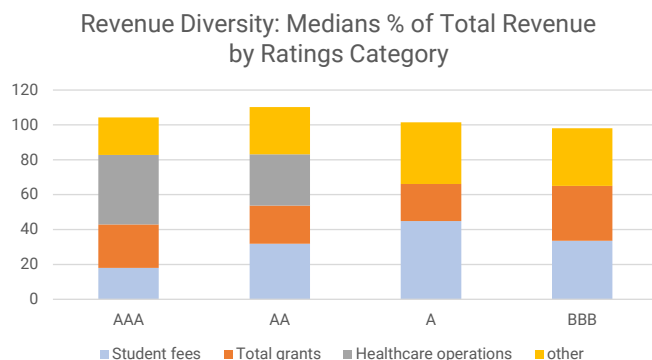
Source: S&P, Morgan Stanley Research

Exhibit 37: Hospital median operating EBITDA margins

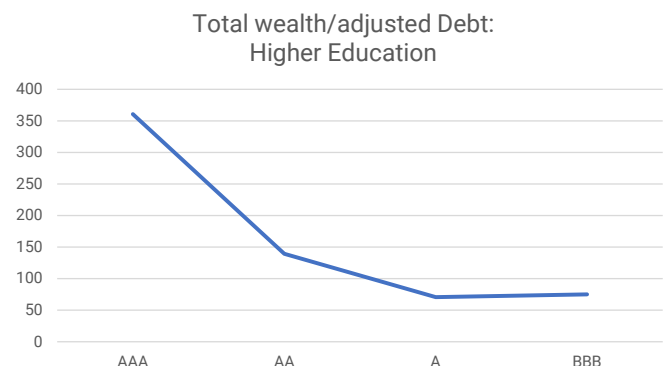
Source: S&P, Morgan Stanley Research

Higher ed: Revenue diversity & asset coverage

For higher education institutions, analysts often look to the mix of revenues. Asset coverage is less important because many endowments are permanently restricted.

Exhibit 38: Revenue diversity matters – less exposure to student fees is typically positive

Source: S&P, Morgan Stanley Research

Exhibit 39: University balance sheets: endowments are often restricted

Source: S&P, Morgan Stanley Research

When Munis Default

Creditors have fewer protections

Government defaults are exceptionally rare — fewer than two per year. When governments do default, creditors may encounter challenges enforcing a judgement.

Not-for-profit and project finance defaults are more common. In some cases, practical constraints limit remedies: the well-being of facility residents or school pupils will always come first.

Most workouts happen outside of bankruptcy

Creditors usually pursue a forbearance agreement to resolve distress. Indentures often provide for a consultant call to recommend business model changes.

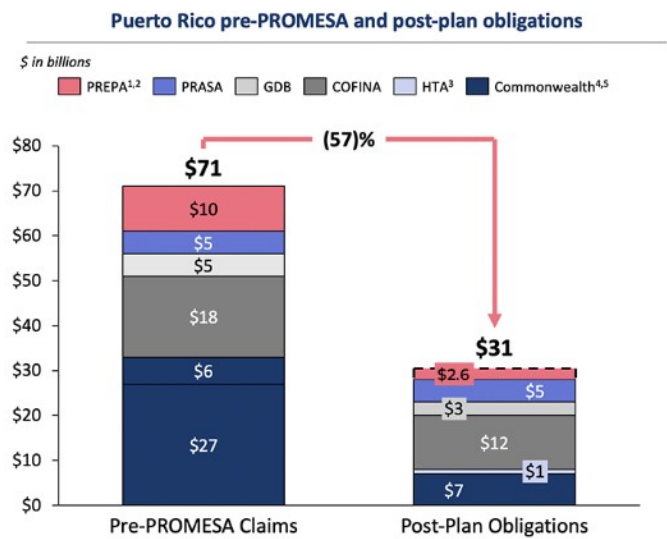
Most covenants and pledges are untested

It is difficult to perfect a lien on taxes or certain kinds of healthcare receivables. Courts vary in their treatment of revenue pledges.

Recoveries vary widely, especially in muni high yield

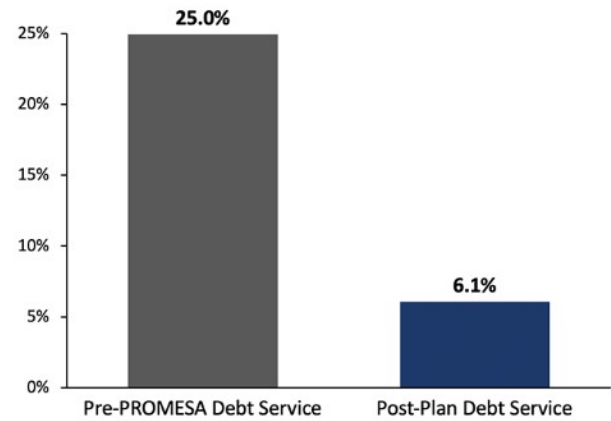
Recoveries are typically lower for untested or discretionary business models. Recoveries can also vary among issuers & pledges.

Puerto Rico Oversight Board: Summary of Restructuring



Puerto Rico pre-PROMESA and post-plan debt service comparison^{6,7}

% of debt policy revenues



1. Pre-PROMESA Claims include ~\$9 billion of bonded debt and ~\$1 million est. GUCs pool per PREPA Title III Amended Disclosure Statement filed on March 3, 2023

2. Post-Plan Obligations of ~\$2.6 billion represents the estimated value PREPA can provide to support creditors pursuant to the Debt Sustainability Analysis included in the PREPA certified 2023 Fiscal Plan, dated June 23, 2023

3. Post-Plan Obligations of ~\$1 billion settles ~\$6 billion of CW related claims under the HTA Plan of Adjustment and supported by HTA toll revenues

4. Pre-PROMESA Claims include GO/PBA of ~\$19 billion, est. GUCs of ~\$3 billion, and ERS of ~\$3 billion, as well as ~\$2 billion PRIFA and ~\$450 million CCDA and MBA (other-clawback entities)

5. Post-Plan Obligations Includes General Obligation claims; but excludes CVI Claims

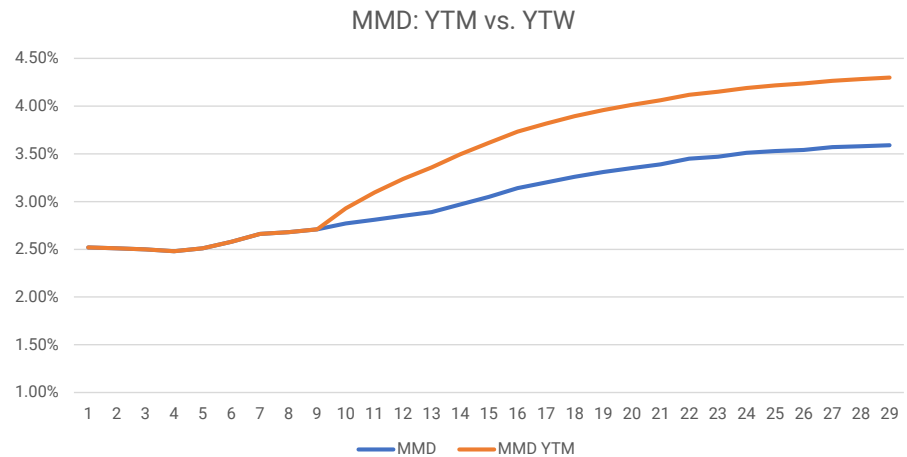
6. Pre-PROMESA Debt Service from 2022 Certified Fiscal Plan and Moody's Investor Service "State Government – U.S. Medians," 2020. (Moody's 2015 number was adjusted to exclude HTA debt service).

7. Post-Plan Debt Service represents FY23 Commonwealth debt service, including COFINA, as a percent of own-source revenues.

Source: Puerto Rico Oversight Board

What Benchmarks Matter

Exhibit 40: Sample MMD yield curve, with yield to worst vs. yield to maturity distinction



Source: MMD, Morgan Stanley Research

Tax-free munis are spread off a AAA, 5% coupon curve

The benchmark curve is an estimated scale for AAA state general obligations. The scale assumes a ten-year call option.

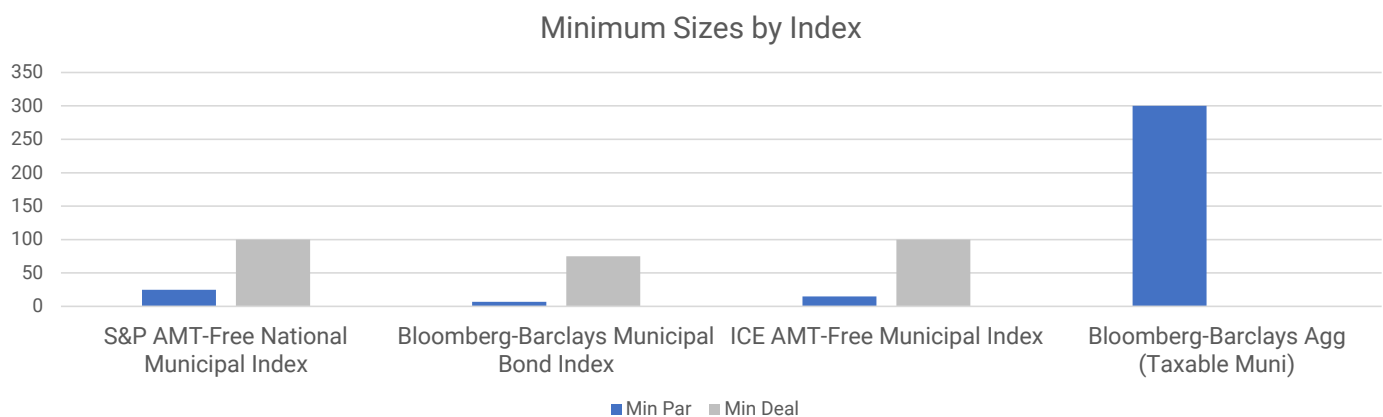
Tax-free munis are not in the agg

Tax-exempt municipal bonds are not included in the Bloomberg-Barclays Aggregate Index.

Some taxable munis are agg-eligible

Taxable munis that meet issue size minimums are index eligible. The agg also includes sufficiently large taxable deals issued by universities and hospitals.

Exhibit 41: Minimum par & deal sizes by muni index

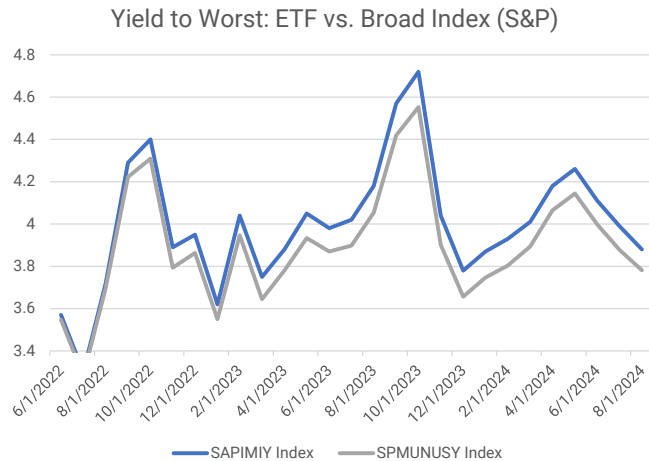


Source: ICE, Bloomberg, MMD, Morgan Stanley Research

For tax-free munis, ETF indices matter more than broad muni indices

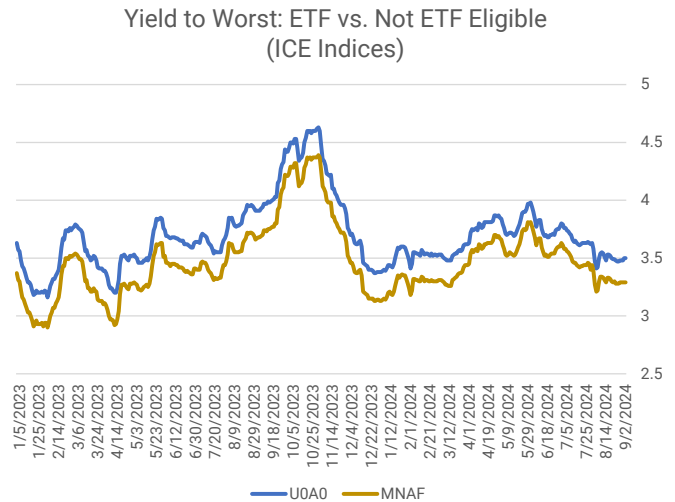
Because there are nearly 1 million municipal CUSIPs outstanding, the broadest muni indices are not replicable. Accordingly, passive municipal ETFs track large-cap indices. These indices exclude healthcare and private universities. They also exclude bonds subject to AMT.

Exhibit 42: Yield to worst (%) – ETF vs. Broad Index



Source: S&P, Morgan Stanley Research

Exhibit 43: Yield to worst (%) – ETF vs. Broad Muni Indices



Source: ICE, Morgan Stanley Research

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(as of August 31, 2024)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1418	38%	338	44%	24%	649	39%
Equal-weight/Hold	1736	47%	351	46%	20%	793	47%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	573	15%	76	10%	13%	230	14%
Total	3,731		765			1673	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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